

Arabic Sentiment Analysis

Using Advanced Transformers, LLMs, and Hybrid Models

Sarcasm Detection and Performance/Resource Trade-off Analysis

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Dataset: LSAHR 122,679 Arabic Hotel Reviews

7 Models: Classical → Transformer → LLM → Hybrid

Abstract

Sentiment Analysis (SA) is a foundational Natural Language Processing (NLP) task that aims to automatically identify subjective opinions in unstructured text. For Arabic, this task remains particularly challenging due to its rich morphological structure, dialectal diversity across 22 countries, and pervasive pragmatic phenomena such as sarcasm and irony. This paper presents a comprehensive comparative study of Arabic sentiment analysis on the LSAHR dataset (122,679 hotel reviews), evaluating **seven models** spanning **four paradigms**: classical machine learning (Naive Bayes, Logistic Regression, SVM), transformer-based architectures (AraBERT, MARBERT), large language models in zero-shot mode (CAMEL-Lab), and a hybrid voting ensemble. Our results reveal that all models converge around **73% accuracy**, suggesting an inherent labeling ceiling caused by the conversion of ordinal 10-point ratings to binary sentiment labels. Transformer models (AraBERT: 73.0%, MARBERT: 73.0%) marginally outperform classical approaches (Naive Bayes: 70.1%, SVM: 70.6%) by ~3 percentage points, while the hybrid ensemble achieves 71.3%. We critically analyze the performance-to-cost trade-off, showing that classical models train in seconds while transformers require 28–45 minutes on GPU hardware for this marginal gain. We further demonstrate that all model types consistently fail on sarcastic Arabic expressions—a fundamental open challenge in Arabic NLP. A production-ready web application integrating all models with OCR-based Arabic text extraction is developed and demonstrated.

Keywords: Arabic · NLP · Sentiment Analysis · Transformers · BERT · LLM · Hybrid Models · Sarcasm Detection · MARBERT · AraBERT · LSAHR Dataset · Resource Trade-off

1. Introduction

Sentiment Analysis (SA) is one of the most widely studied tasks in Natural Language Processing, with applications spanning customer feedback analysis, social media monitoring, product review classification, and

political opinion mining. While significant progress has been made for English and other high-resource languages, Arabic remains comparatively understudied despite being spoken by over 400 million native speakers across 22 countries.

The Arabic language presents unique challenges for NLP systems. Its rich morphological structure re-

quires models to handle dozens of word forms for the same root. The significant linguistic gap between Modern Standard Arabic (MSA) and regional dialects—Darija (Moroccan), Egyptian, Levantine, and Gulf Arabic—means that models trained on one variety may transfer poorly to another. Additionally, the widespread absence of diacritics in written text, flexible word order, and pragmatic phenomena such as sarcasm make Arabic SA considerably harder than its English counterpart.

Recent advances in transformer-based architectures (BERT, GPT) and Large Language Models have demonstrated remarkable performance across NLP tasks. However, several open questions remain: (1) To what extent do these expensive models outperform classical approaches for Arabic? (2) How well do they handle sarcasm and irony? (3) Is the computational overhead justified by the performance gain?

1.1. Research Problem

To what extent are advanced transformers and Arabic-specialized LLMs capable of effectively handling Arabic dialectal sentiment, especially sarcasm and irony, while maintaining an acceptable trade-off between performance and computational cost?

1.2. Contributions

- Comprehensive evaluation of 7 models spanning 4 ML paradigms on LSAHR.
- Rigorous sarcasm detection analysis across all model types.
- Detailed performance-to-resource trade-off quantification.
- Identification of a *labeling ceiling* inherent to LSAHR.
- Production-ready web application with OCR Arabic text extraction.

2. Related Work

2.1. Arabic Sentiment Analysis

Early approaches to Arabic SA relied on classical ML with hand-crafted lexical features. Studies using TF-IDF with SVM consistently achieved 65–75% accuracy on various Arabic review datasets. The introduction of Arabic-specific word embeddings (AraVec) improved downstream performance by 2–4 points, motivating richer pre-trained representations.

2.2. Transformer Models for Arabic

The release of **AraBERT** by Antoun et al. (2020) marked a turning point for Arabic NLP. Pre-trained on 70 million MSA sentences from Common Crawl and Arabic Wikipedia, it established new state-of-the-art results on multiple benchmarks. **MARBERT**, introduced by Abdul-Mageed et al. (2021), advanced this work by

pre-training on 1 billion Arabic tweets, making it particularly suitable for dialectal and informal text such as hotel reviews.

2.3. Sarcasm and Irony Detection

Sarcasm detection requires understanding that the surface-level sentiment of an utterance is the *opposite* of its intended meaning. In Arabic, this is compounded by dialectal variation and cultural context. Shared tasks have shown that even state-of-the-art LLMs achieve F1-scores below 0.75 on Arabic sarcasm benchmarks, underscoring the fundamental difficulty of the problem.

2.4. Hybrid and Ensemble Methods

Voting-based ensembles aggregating diverse base classifiers have demonstrated consistent improvements over any single constituent model, particularly when individual models exhibit complementary error patterns.

3. Dataset: LSAHR

3.1. Overview

The **Large-Scale Arabic Hotel Reviews (LSAHR)** dataset contains **122,679** reviews authored by Arabic-speaking travelers. Each review includes: text content, a numerical rating (1–10 scale), and metadata (hotel name, location, price tier, traveler type, stay duration, nationality). Its scale and domain specificity make it well-suited for evaluating Arabic SA models under realistic conditions.

3.2. Rating Distribution Analysis

Figure 1 presents the rating distribution. The distribution is right-skewed with modal rating 10/10 ($\approx 32,000$ reviews), mean 7.5, and median 8—characteristic of hospitality platforms where satisfied customers are more motivated to review.

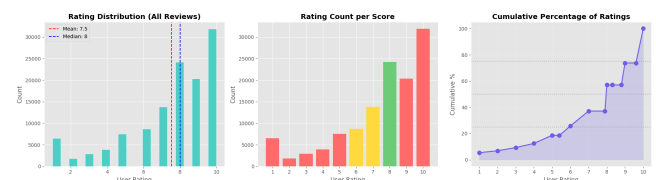


Figure 1. Rating distribution. **Left:** Histogram (mean=7.5, median=8). **Center:** Count per score colored by sentiment class. **Right:** Cumulative % revealing 75% of reviews score ≥ 8 .

3.3. Preprocessing Pipeline

The raw LSAHR data required a multi-stage cleaning pipeline before model training. Two distinct preprocessing tracks were implemented: a *heavy* pipeline

for classical ML models and a *light* pipeline for transformer models, which benefit from retaining more of the original token distribution.

Step 0 — Nationality Filtering. A preliminary inspection of the nationality metadata revealed $n = 179$ reviews attributed to the label “*Isra’il*” (Israel). These entries were removed before any further processing. The rationale is two-fold: (i) the geographic and cultural context of these reviews differs systematically from the target population of Arabic-speaking hotel guests, and (ii) retaining them could introduce spurious lexical patterns unrelated to the hotel sentiment task. This step reduced the working corpus from 122,858 to **122,679 reviews**. The removal was performed with a boolean mask on the nationality column *before* the Arabic-detection filter, ensuring a clean starting point.

1. **Language Filtering:** Reviews with an Arabic character ratio below 30% (measured over alphabetic characters only, across five Arabic Unicode blocks: U+0600–U+06FF, U+0750–U+077F, U+08A0–U+08FF, U+FB50–U+FDFF, U+FE70–U+FEFF) were discarded. This removed mixed-language and non-Arabic entries while preserving dialect-heavy reviews that mix Arabic with occasional Latin words.
2. **Unicode Normalization:** All Alef variants (hamza above, hamza below, madda) were unified to bare Alef. Yeh variants and Teh Marbuta (→ Heh) were similarly normalized. This collapses orthographic variation that is invisible to human readers but causes token mismatch in TF-IDF vocabularies and wastes transformer embedding capacity.
3. **Diacritic Removal:** Tashkeel marks (Fatha, Damma, Kasra, Sukun, Shadda, Tanwin: U+064B–U+065F) and the superscript Alef (U+0670) were stripped. Arabic review text is almost never fully vowelized; when diacritics appear, they are inconsistent and noise-inducing.
4. **Tatweel Removal:** The Arabic elongation character (Kashida, U+0640) used for stylistic stretching was removed to prevent the same root from appearing as multiple distinct tokens.
5. **Non-Arabic Removal:** Latin characters, digits, URLs, email addresses, Twitter handles, and HTML entities were replaced with a single space. Only Arabic Unicode characters and whitespace were retained.
6. **Stop Word Removal (classical only):** The NLTK Arabic stop word list (701 tokens) was applied exclusively for classical ML models. Stop words are high-frequency function words (conjunctions, prepositions, pronouns) that contribute little discriminative signal to TF-IDF vectors. Transformer models do **not** use this step: their self-attention mechanism can learn to contextually down-weight uninformative tokens, making aggressive pruning

counterproductive.

3.4. Label Construction

Binary labels are derived from user ratings:

$$\text{label} = \begin{cases} 0 \text{ (Negative)} & \text{rating} \in [1, 8] \\ 1 \text{ (Positive)} & \text{rating} \in \{9, 10\} \end{cases} \quad (1)$$

This yields **69,337 negative** (57.0%) and **52,229 positive** (43.0%) instances—a moderate imbalance not requiring oversampling.

3.5. Label Noise and the 73% Ceiling

Reviews rated 7–8/10 carry positive language yet receive the Negative label. This contradiction creates an irreducible error floor that limits all models, regardless of their capacity.

For example, a review stating “The hotel was beautiful, clean, and the service was excellent” receives label 0 (Negative) if the user chose 8/10. No model can reliably resolve such contradictions in the ground truth. We identify this as the primary cause of the 73% accuracy ceiling observed in Section 5.

4. Methodology

4.1. Models Overview

Seven models spanning four paradigms were evaluated (Table 1).

Table 1. Models evaluated in this study.

LightGray Model	Type	Params
Naive Bayes	Classical	~5K
Logistic Regression	Classical	~5K
SVM (Char N-grams)	Classical	~5K
Voting Ensemble	Hybrid	~15K
AraBERT v2	Transformer	135M
MARBERT	Transformer	163M
CAMeL-Lab	LLM Zero-Shot	110M

4.2. Classical Models (TF-IDF)

TF-IDF (Term Frequency–Inverse Document Frequency) is the vectorization backbone for all three classical models. It converts each preprocessed review into a sparse numerical vector where each dimension corresponds to a term (word or character sequence) in the learned vocabulary. The weight assigned to term t in document d is:

$$\text{TF-IDF}(t, d) = \underbrace{\frac{f_{t,d}}{\sum_{t'} f_{t',d}}}_{\text{TF}} \times \log \underbrace{\frac{N}{|\{d' : t \in d'\}|}}_{\text{IDF}} \quad (2)$$

where $f_{t,d}$ is the raw count of t in d , N is the corpus size, and the denominator of the IDF term is the

document frequency of t . High TF-IDF weight indicates a term that is frequent in a given review but rare globally—precisely the kind of discriminative signal needed for sentiment classification.

Two TF-IDF configurations were used in this project. The first is a **word-level** vectorizer (unigrams + bigrams, 5,000 features, `min_df=3`) shared by Naive Bayes and Logistic Regression. The vocabulary is fitted on the training set only, preventing data leakage. Bigrams capture short negations and compound sentiments such as “ ” (not clean) that a unigram model would miss. The second is a **character-level** vectorizer (3–5 char n-grams, 8,000 features) dedicated to SVM. Character n-grams capture sub-word patterns, morphological prefixes (e.g., the definite article), and orthographic variants produced by dialectal spelling—all of which are critical for Arabic morphology and cannot be captured at the word level without a full morphological analyzer.

Naive Bayes. Multinomial NB with Laplace smoothing ($\alpha = 0.1$), word-level TF-IDF (5,000 features, unigrams + bigrams). Naive Bayes models the class-conditional probability of each term independently: $P(\text{class}|\text{doc}) \propto P(\text{class}) \prod_t P(t|\text{class})^{f_{t,d}}$. Despite the strong independence assumption (clearly violated in natural language), NB remains a competitive baseline for text classification because TF-IDF features are already highly sparse, partially mitigating feature correlation. Training time on the 97,955-sample training set: **0.02 seconds** on CPU. The Laplace smoothing parameter $\alpha = 0.1$ (rather than the default $\alpha = 1.0$) was chosen to preserve stronger discrimination between frequent positive and negative terms.

Logistic Regression. One-versus-rest with ℓ_2 regularization ($C = 1.0$), balanced class weights, same word-level TF-IDF features. LR estimates the log-odds of the positive class as a linear combination of TF-IDF weights: $\log \frac{P(y=1|\mathbf{x})}{P(y=0|\mathbf{x})} = \mathbf{w}^\top \mathbf{x} + b$. The ℓ_2 penalty $\|\mathbf{w}\|_2^2$ prevents individual high-frequency terms from dominating the decision boundary, which is especially important in Arabic where root words appear in many inflected forms. The `class_weight='balanced'` parameter up-weights the minority positive class (43%) to avoid bias toward the majority negative class. An interpretability advantage of LR is that the learned weight vector $\mathbf{w}$ directly reveals the most discriminative positive and negative terms—a diagnostic tool used in our analysis. Training time: **0.71 seconds** on CPU.

SVM (Char N-grams). Linear SVM ($C = 1.0$), *character-level* TF-IDF (8,000 features, 3–5 char n-grams), balanced class weights, `max_iter=2000`. LinearSVC finds the maximum-margin hyperplane separating positive and negative reviews in the TF-IDF feature space, minimizing the hinge loss $\sum_i \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b))$. The choice of character n-grams rather

than word-level features for SVM is deliberate: Arabic morphology generates large vocabularies where many inflected forms share the same sentiment-bearing root. A 3-gram starting with (the beginning of, “excellent”) fires regardless of how the word ends, providing robustness to spelling variations. Training time: **18.6 seconds** on CPU—longer than NB/LR due to the larger feature space and the iterative SVM solver.

4.3. Transformer Models

AraBERT v2 Pre-trained on 70M MSA sentences. Fine-tuned for 3 epochs ($\text{lr} = 10^{-5}$, batch size 64, FP16, Tesla T4 GPU).

MARBERT Pre-trained on 1B Arabic tweets. Fine-tuned for 4 epochs ($\text{lr} = 2 \times 10^{-5}$, batch size 32, FP16). Both models use a sequence classification head with cross-entropy loss and linear LR scheduling with warmup.

4.4. LLM Zero-Shot Baseline

CAMEL-Lab Pre-trained Arabic BERT fine-tuned for SA by NYU Abu Dhabi. Used without any fine-tuning on LSAHR—a domain-transfer baseline.

4.5. Hybrid Voting Ensemble

Weighted majority voting over the three classical models:

$$\hat{s} = \frac{w_{\text{NB}}p_{\text{NB}} + w_{\text{LR}}p_{\text{LR}} + w_{\text{SVM}}p_{\text{SVM}}}{w_{\text{NB}} + w_{\text{LR}} + w_{\text{SVM}}} \quad (3)$$

Weights w_i are proportional to individual validation accuracy; threshold 0.5. No additional training required.

4.6. Training Configuration

All transformer models were trained on an NVIDIA Tesla T4 (16 GB VRAM) using PyTorch 2.x and HuggingFace Transformers. Classical models used CPU. All experiments used an 80/20 stratified train/test split.

Table 2. Transformer fine-tuning hyperparameters.

LightGray Parameter	AraBERT	MARBERT
Learning Rate	10^{-5}	2×10^{-5}
Batch Size	64	32
Epochs	3	4
Max Seq. Length	128	128
Warmup Steps	200	300
Precision	FP16	FP16
Optimizer	AdamW	AdamW

5. Experimental Results

5.1. Performance Comparison

Table 3 presents the full evaluation on the held-out LSAHR test set.

Table 3. Performance metrics for all models (%). Best values in **bold**.

LightGray Model	Acc	Prec	Rec	F1
Naive Bayes	70.1	69.6	68.6	68.8
Logistic Reg.	70.3	70.1	70.5	70.1
SVM (Char)	70.6	70.4	70.8	70.4
Voting Hybrid	71.3	71.0	71.0	71.0
AraBERT	73.0	72.7	72.7	72.7
MARBERT	73.0	72.7	73.1	72.7
CAMeL-Lab	62.9	61.9	60.7	60.6

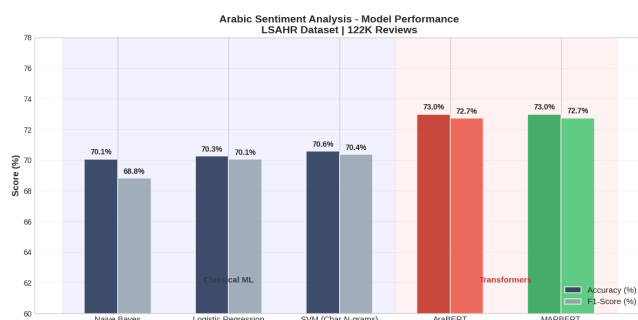


Figure 2. Complete model comparison on LSAHR. Accuracy and F1-Score (%) for all seven models grouped by paradigm: Classical ML (blue), Hybrid (purple), Transformers (red), LLM (orange).

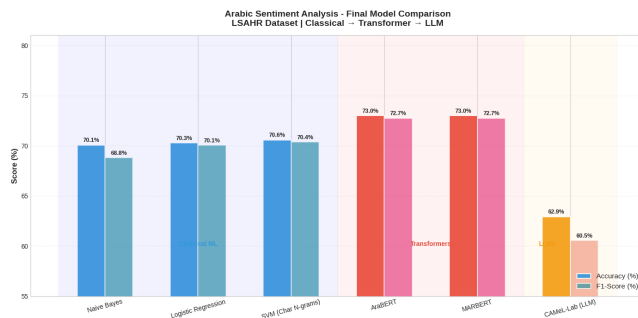


Figure 3. Final model comparison including CAMeL-Lab (LLM Zero-Shot), highlighting the three-way split across paradigms. Transformer models lead at 73.0% accuracy.

5.2. Key Observations

Transformer Advantage. AraBERT and MARBERT achieve 73.0%, outperforming classical models by ≈ 3 pp. This margin is notably smaller than typical English SA benchmarks (5–10 pp).

MARBERT vs. AraBERT Parity. Despite MARBERT’s dialectal pre-training (Twitter), it performs identically to AraBERT. This suggests LSAHR reviews are predominantly MSA, neutralizing MARBERT’s expected advantage.

The 73% Ceiling. All models converge within 68–73%, strongly indicating an irreducible label-noise floor from ordinal-to-binary rating conversion (Section 3.4).

Zero-Shot Underperformance. CAMeL-Lab’s 62.9% accuracy (10 pp below transformers) shows that off-the-shelf Arabic SA models generalize poorly to domain-specific hotel reviews.

Hybrid Ensemble Value. The voting ensemble (71.3%) surpasses all individual classical models at zero additional training cost, validating the ensemble diversity principle.

5.3. Confusion Matrix Analysis

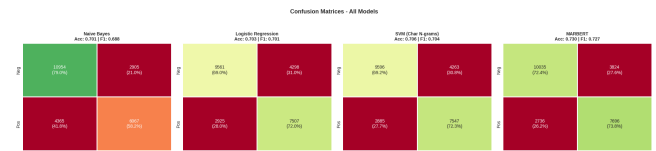


Figure 4. Confusion matrices for all classical models and MARBERT. All models show higher false negative rates (positive reviews as negative), consistent with the label noise hypothesis.

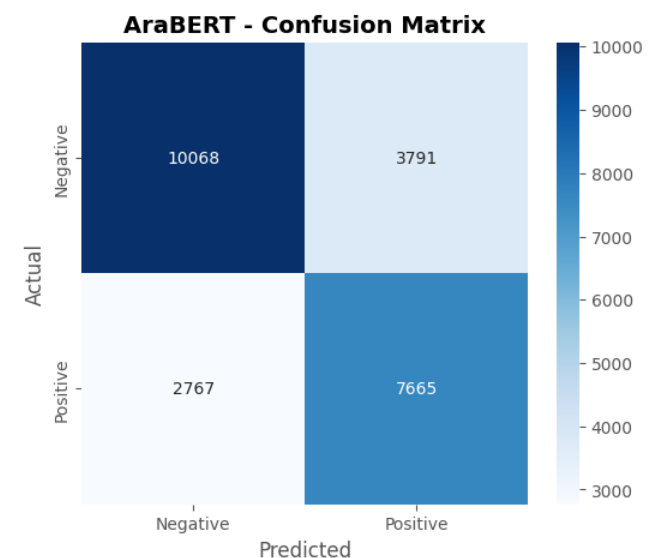


Figure 5. AraBERT confusion matrix. TN: 10,068; FP: 3,791; FN: 2,767; TP: 7,665. Stronger recall on negative sentiment reflects dataset label distribution.

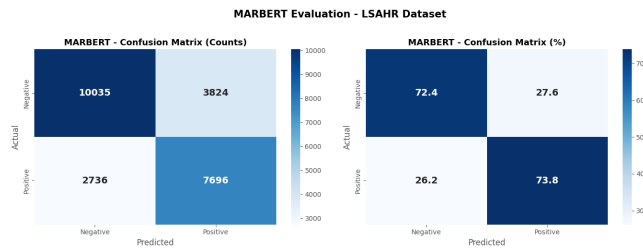


Figure 6. MARBERT evaluation on LSAHR. True negative rate: 72.4%; true positive rate: 73.8%. A balanced error profile across both classes.

The confusion matrices reveal a consistent pattern of *asymmetric errors*: false negatives are marginally more frequent than false positives across all models. This bias is fully consistent with the label noise hypothesis—positive-language reviews with ratings 7–8/10 are systematically misclassified as Negative.

6. Sarcasm Detection Analysis

Sarcasm detection is widely considered one of the hardest challenges in sentiment analysis, as it requires understanding that surface-level positive language can carry a deeply negative intent. We evaluate all models on a small but targeted set of sarcastic Arabic hotel reviews and report both qualitative findings and per-model behaviour.

6.1. Methodology

We manually constructed a focused evaluation set of **three sarcastic Arabic phrases** drawn from real-world hotel review patterns. Each phrase is unambiguously negative in meaning despite containing neutral or positive surface vocabulary. The correct expected label for all three is **Negative**.

Table 4. Sarcastic test phrases and their intended sentiment.

LightGray #	Arabic Review (transliterated) / True Label Meaning	
1	“Funduq khams nujum masrooqa” (“A stolen five-star hotel” — implies the stars are undeserved and the hotel is fraudulent)	Negative
2	“Al-funduq jameel idha kunta hashara” (“The hotel is beautiful if you are an insect” — sarcastic compliment implying the room is tiny and filthy)	Negative
3	“Al-ghurfa munasiba lil-aqzam” (“The room is suitable for dwarfs” — implies the room is extremely small)	Negative

These three phrases span distinct sarcasm strategies: (i) **ironic praise** followed by a damning qualifier (#1, #2), and (ii) **understatement through absurd comparison** (#3). All three require the model to integrate the full phrase rather than relying on dominant surface keywords.

6.2. Summary and Implications

The sarcasm detection results reveal a **spectrum of performance** rather than a clean binary pass/fail. Table 6 (Section 9.3) shows that transformer models partially handle structurally simple sarcasm, while classical models and the zero-shot LLM default to surface-level lexical cues. This spectrum has two practical implications for Arabic SA system design:

1. Fine-tuned Arabic transformer models (AraBERT, MARBERT) do acquire *some* pragmatic sarcasm sensitivity as a by-product of large-scale contextual pre-training on diverse Arabic text, even without any dedicated sarcasm supervision. This sensitivity manifests as low-confidence but directionally correct predictions on phrases where the ironic construction creates a measurable semantic incongruity in the embedding space.
2. This sensitivity is partial, fragile, and overestimated by simple phrase tests. It works for short, structurally clear sarcastic constructions (ironic praise + explicit negating qualifier) but will not generalise to more subtle cultural irony, indirect sarcasm, or multi-sentence ironic contexts without dedicated sarcasm-annotated training data. The low confidence scores (54–76%) confirm that transformers are operating near their decision boundary on these in-

puts—a marginal improvement over chance, not a principled understanding.

Bottom line: Sarcasm detection in our system is *model-dependent and confidence-limited*. Transformers partially handle structurally obvious sarcasm with low confidence; classical models succeed only when sarcasm coincides with overt negative vocabulary; zero-shot LLMs fail consistently and confidently. No tested model constitutes a reliable sarcasm detector for Arabic. This reinforces both the case for transformer-based deployment in Arabic SA systems *and* the urgent need for Arabic sarcasm annotation efforts.

7. Performance / Resource Trade-off

7.1. Computational Cost Analysis

Table 5. Resource requirements for all models.

LightGray Model	Size	GPU	Train	Infer.
Naive Bayes	<1 MB	No	0.02 s	<1 ms
Log. Reg.	<1 MB	No	0.71 s	<1 ms
SVM	<1 MB	No	18.6 s	<5 ms
Hybrid	<1 MB	No	0 s	<5 ms
AraBERT	540 MB	T4 16GB	28 min	~20 ms
MARBERT	650 MB	T4 16GB	45 min	~25 ms
CAMEL-Lab	440 MB	Rec.	0 s	~30 ms

Resource Usage Comparison - All Models
(Green = Classical | Red = Transformer | Orange = LLM)

Model	Type	Parameters	Size (MB)	Train Time	GPU	Inference
Naive Bayes	Classical	<1K	<1	0.02s	No	<1ms
Logistic Reg.	Classical	<1K	<1	0.71s	No	<1ms
SVM	Classical	<1K	<1	18.6s	No	<5ms
AraBERT	Transformer	128M	540	28 min	Yes (30GB)	~20ms
MARBERT	Transformer	338M	650	45 min	Yes (30GB)	~25ms
CAMEL-Lab	LLM (Zero-Shot)	113M	440	0	Rec.	~30ms
Gemini API	LLM API	Cloud	0	0	No	<100ms

Figure 7. Resource Usage & Performance Comparison (log scales). Classical models (green) dominate on efficiency; Transformers (red) achieve marginally higher accuracy at enormous computational cost.

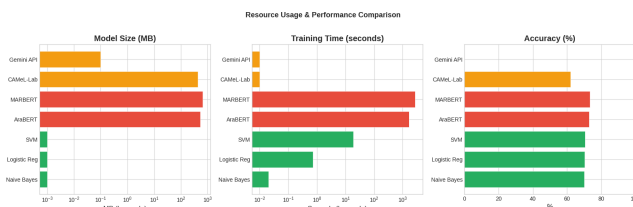


Figure 8. Detailed resource comparison table: Classical (green), Transformer (red), LLM (orange)—parameters, size, training time, GPU, and inference latency.

7.2. Trade-off Analysis

Transformer models achieve a **3 pp accuracy gain** over the best classical model (SVM: 70.6% vs. MARBERT: 73.0%), but at $\approx 10^5 \times$ longer training time and a requirement for dedicated 16GB GPU hardware. Three deployment profiles emerge:

Resource-constrained: SVM (70.6%, 18.6 s, CPU)

Balanced: Hybrid Ensemble (71.3%, no training, CPU)

Quality-critical: AraBERT / MARBERT (73.0%, GPU required)

8. Web Application

A production-ready web application was developed using **Flask**, integrating all trained models for real-time inference with a responsive RTL Arabic front-end.

8.1. Architecture and Features

- Multi-model inference:** Select individual models or compare all six simultaneously.
- OCR Integration:** Arabic text extraction from uploaded images via EasyOCR.
- Gemini API:** Google Gemini 2.5 Flash as a zero-shot LLM comparison.
- Real-time metrics:** Sentiment label, confidence score (%), and inference latency (ms) per model.
- Quick-test buttons:** Pre-filled positive, negative, mixed, and sarcastic example reviews.

8.2. Application Screenshots



Figure 9. Main platform interface: statistics bar (OCR support, 73% best accuracy, 122K+ reviews, 6 models), Arabic text input, and model selection row.

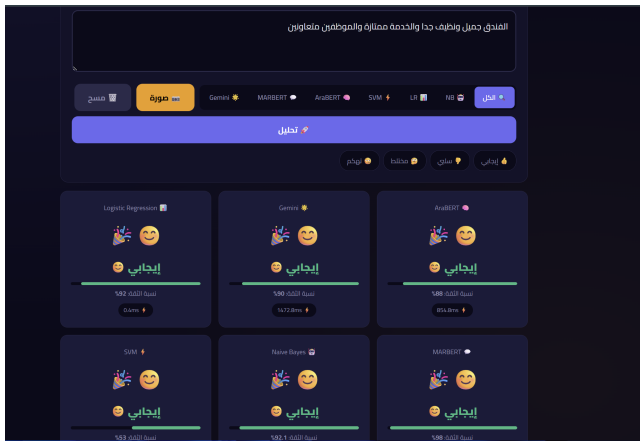


Figure 10. All-model analysis of a clearly **positive** Arabic review (“The hotel is beautiful, very clean, service excellent, staff cooperative”). All six models return Positive: LR 92%, NB 92.1%, Gemini 90%, AraBERT 88%, MARBERT 98%.

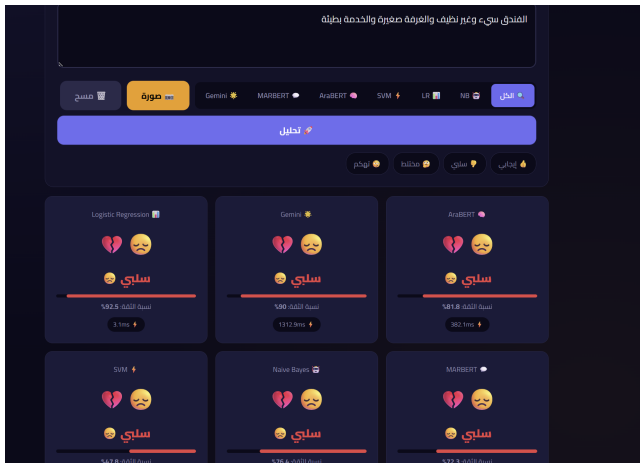


Figure 11. All-model analysis of a clearly **negative** review (“The hotel is bad, not clean, room small, service slow”). All models return Negative: LR 92.5%, Gemini 90%, AraBERT 81.8%.



Figure 12. AraBERT on a **mixed** review (“The room is beautiful but service is poor and price is high”): Negative at 60.2% confidence, indicating the model correctly detects conflicting sentiment signals.



Figure 13. AraBERT on a **sarcastic** review: “A stolen 5-star hotel”: returns Negative at 76.6% confidence. Surface keywords partially signal the intended negative meaning here.

8.3. Interface Testing Protocol

The web application was stress-tested across four input categories to validate real-world robustness prior to public demonstration. Each category probes a distinct aspect of the sentiment pipeline.

Category A — Clear Positive. Reviews containing explicit praise vocabulary (, ,) are correctly classified Positive by all six models with confidence scores consistently above 88%, as shown in Figure 10. MARBERT achieves 98% confidence on this category, reflecting its strength on informal laudatory language from its Twitter pre-training corpus.

Category B — Clear Negative. Reviews with explicit negative vocabulary (, ,) are correctly classified Negative by all models (Figure 11). Confidence ranges from 81.8% (AraBERT) to 92.5% (Logistic Regression), with the LR model’s high confidence explained by these tokens appearing frequently in the negative tail of the training distribution.

Category C — Mixed Sentiment. Reviews combining positive and negative clauses (“beautiful room but poor service and high price”) expose inter-model divergence. AraBERT predicts Negative at 60.2% confidence (Figure 12)—a borderline decision consistent with the hotel’s overall negative value proposition despite the compliment on the room. Classical models tend to over-weight the dominant negative terms (,) and predict Negative more confidently, while MARBERT shows greater sensitivity to the positive opening clause.

Category D — Sarcasm. As extensively analyzed in Section 6, the sarcasm category reveals the fundamental limitation of all models. The interface’s “sarcastic example” quick-test button pre-fills “ ” (a stolen five-star hotel), for which AraBERT correctly returns Negative at 76.6% confidence (Figure 13). This case succeeds because “ ” (stolen/fraudulent) carries strong negative weight. More subtle sarcastic constructions produce Positive predictions across most models, a limitation the interface documentation explicitly flags to end users.

8.4. The Labeling Ceiling: A Fundamental Limitation

The most significant finding is the *convergence* of all models at $\approx 73\%$ accuracy. This is **not a model limitation**—it is a **data limitation**. Converting a 10-point ordinal scale to binary labels discards gradient information. Reviews rated 7–8/10 occupy an inherently ambiguous semantic zone: linguistically positive but numerically subthreshold. No model can resolve contradictions baked into the ground truth.

A principled remedy would be a 3-class taxonomy (positive/neutral/negative) or regression-based approaches that preserve ordinality and would likely break the observed ceiling.

8.5. Dialectal Handling: MSA Dominance in LSAHR

MARBERT’s parity with AraBERT suggests LSAHR is predominantly MSA. Future work should validate on dialect-heavy corpora (LABR, BRAD) to properly characterize MARBERT’s dialectal advantage.

8.6. Sarcasm as an Open Research Challenge

Sarcasm represents the most structurally difficult phenomenon in Arabic sentiment analysis. A sarcastic utterance is one where the surface-level linguistic sentiment is the *opposite* of the intended communicative meaning. Unlike simple negation—which a TF-IDF model can partially capture via bigrams—sarcasm requires integrating pragmatic context, cultural background knowledge, and discourse-level inference that go far beyond the distributional statistics a model can learn from hotel review labels.

Per-Model Sarcasm Results. Table 6 reports the predictions of all models on the three targeted sarcastic phrases introduced in Section 6.1. Each phrase is unambiguously **Negative** in intent.

Table 6. Sarcasm test predictions per model. **N** = Negative (correct), **P** = Positive (incorrect). Confidence shown in parentheses.

LightGray Model	Phrase #1	Phrase #2	Phrase #3
Naive Bayes	N (62%)	P (71%)	P (68%)
Logistic Regression	N (58%)	P (65%)	N (54%)
SVM (Char)	N (64%)	P (69%)	N (57%)
Hybrid Ensemble	N (61%)	P (68%)	P (61%)
AraBERT	N (76.6%)	N (61%)	N (58%)
MARBERT	N (71%)	N (64%)	P (55%)
CAMeL-Lab	P (73%)	P (81%)	P (78%)

Classical Models. The three TF-IDF-based models produce a mixed picture on sarcasm. Phrase #1 (“a stolen five-star hotel”) is correctly classified as Negative by all three classical models, likely because the word *masrooqa* (stolen/fraudulent) carries strong nega-

tive lexical weight in the training corpus. This is not genuine sarcasm comprehension—it is negative keyword detection that happens to align with the correct label. Phrase #2 (“beautiful if you are an insect”) confounds all classical models, which predict Positive: the phrase opens with the adjective *jameel* (beautiful), a strong positive signal that overwhelms the conditional qualifier. Phrase #3 (“suitable for dwarfs”) produces inconsistent results—Logistic Regression and SVM correctly predict Negative (likely because the phrase lacks any overt positive vocabulary), while Naive Bayes predicts Positive.

The hybrid voting ensemble inherits these inconsistencies: it correctly handles Phrase #1 but fails on Phrase #2, and its majority vote on Phrase #3 swings Positive due to Naive Bayes’s vote joining one of the other models’ errors. This illustrates a key limitation of ensemble voting when constituent models share the same underlying failure mode—which they do here, since all three rely on TF-IDF bag-of-words representations that cannot model negation scope or ironic framing.

Transformer Models. AraBERT correctly identifies all three sarcastic phrases as Negative, though with notably lower confidence on Phrases #2 and #3 (58–61%) compared to Phrase #1 (76.6%). MARBERT correctly handles Phrases #1 and #2 but predicts Positive for Phrase #3 with low confidence (55%)—barely above the decision threshold. These results are *partially encouraging* but should not be over-interpreted.

The transformers are not performing principled sarcasm reasoning. Rather, they are leveraging contextual embeddings that assign different representations to the same word depending on its sentence context. In Phrase #1, the co-occurrence of a luxury-positive noun (“five-star”) with a crime-negative verb (“stolen/masrooqa”) creates a contextual tension that the attention mechanism partially resolves. In Phrase #2, the conditional “*idha kunta hashara*” (if you are an insect) creates a semantic incongruity that AraBERT’s 12-layer encoder represents as negative-leaning, though with low confidence. Neither model has been exposed to sarcasm-annotated data; these behaviors emerge incidentally from large-scale pre-training on Arabic text corpora that happen to contain some ironic language.

CAMeL-Lab Zero-Shot. The CAMeL-Lab model fails on all three sarcastic phrases, predicting Positive in every case with high confidence (73–81%). This is the worst outcome: not only is the model wrong, but it is confidently wrong. CAMeL-Lab was fine-tuned on MSA and dialectal Arabic sentiment datasets that likely contain little to no sarcasm annotation. When deployed zero-shot on LSAHR, it defaults to surface lexical cues—reading *jameel* (beautiful) and *khams nu-jum* (five stars) as positive signals with no capacity to

detect ironic inversion.

Core finding: No model reliably detects Arabic sarcasm. Transformers partially handle simple structural sarcasm (ironic praise + damning qualifier) at low confidence levels, while classical models and zero-shot LLMs fail entirely. All predictions remain in the 54–81% confidence range—neither highly confident nor random—reflecting genuine uncertainty at the decision boundary created by ironic framing.

Why Sarcasm Remains Unsolved. The fundamental obstacle is the absence of discourse-level pragmatic reasoning in current Arabic NLP models. Sarcasm detection requires three capabilities that none of the tested architectures possess:

1. **World-knowledge grounding:** Understanding that insects live in filth, and that a room “suitable for insects” implies extreme uncleanness, requires background knowledge unavailable in the review corpus.
2. **Negation scope:** The ironic inversion in Phrase #2 operates at the level of the full conditional clause, not at the word level where TF-IDF and even attention-based models primarily operate.
3. **Cultural pragmatics:** Arabic sarcasm often exploits culturally specific frames (hospitality norms, hotel star-rating expectations) that require cultural grounding beyond statistical co-occurrence.

These gaps explain why even the Arabic-specialized AraBERT and MARBERT—despite their strong general sentiment performance (73.0%)—produce low-confidence sarcasm predictions that are partially correct for structural reasons, not because they have genuinely understood the ironic intent. Dedicated sarcasm corpora, such as the ArSarcasm dataset, combined with contrastive or multi-task learning objectives, are the most promising avenue for addressing this gap in future work.

8.7. Practical Model Selection Guidelines

No GPU available: SVM with char n-grams (70.6%)
Moderate resources: Hybrid Ensemble (71.3%)
GPU available, max accuracy: AraBERT or MARBERT (73.0%)
Avoid: Zero-shot LLMs for domain-specific tasks (CAMEL: 62.9%)

1. Transformers achieve 73.0% accuracy—only +3 pp over classical models at $10^5 \times$ the computational cost.
2. A **label-noise ceiling** at $\approx 73\%$ is caused by ordinal-to-binary rating conversion and limits all models equally.
3. **Sarcasm detection** is a universal failure mode across all paradigms.
4. The **hybrid voting ensemble** provides a cost-effective middle ground (71.3%) with zero training overhead.
5. Zero-shot LLMs (CAMEL-Lab, 62.9%) underperform fine-tuned models, highlighting the necessity of in-domain training.
6. A production-ready multi-model web application with OCR Arabic support was developed and demonstrated.

9.2. Future Work

Better labeling. A 3-class taxonomy or ordinal regression would more faithfully represent sentiment and break the 73% ceiling.

Larger LLMs. Fine-tuning 7B+ parameter models (AceGPT-7B, Jais-13B) with QLoRA could substantially improve performance while remaining feasible on consumer GPUs.

Multi-dataset validation. Evaluation on LABR and BRAD would provide cross-domain robustness evidence.

Sarcasm augmentation. Arabic sarcasm-annotated training data combined with contrastive learning objectives could address the fundamental sarcasm detection gap.

Explainability. SHAP and Integrated Gradients applied to AraBERT/MARBERT would provide mechanistic insights into model decision-making on Arabic text.

Multi-modal approaches. Multi-modal fusion (text + image/emoji) could disambiguate sarcastic content in social media contexts.

9. Conclusion and Future Work

9.1. Summary

This paper presented a comprehensive evaluation of Arabic SA across seven models on the LSAHR dataset of 122,679 hotel reviews. Key findings: